

DAW_P12_Andreev — LDAP

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Practice: 12. LDAP

Environment: Arch Linux host in VS Code + Ubuntu Server VM

Introduction

This report documents the deployment of an LDAP directory service on Ubuntu Server and its integration with Apache for centralized authentication.

1. LDAP installation, service enablement, and firewall

1.0 Environment notes

VM IP: 192.168.1.140

Hostname: mtu

SSH access from Arch Linux: `ssh user@192.168.1.140` (or replace `user` with actual Ubuntu username)

Notes: Ubuntu 24.04.3 LTS on VirtualBox. SSH service enabled and running.
Bonus: will implement StartTLS or LDAPS for secure LDAP connections.

1.1 System update

What I did: Updated package lists and checked system upgrades on Ubuntu Server.

Commands used: `sudo apt update && sudo apt upgrade -y`

Result: The system repositories were reached successfully, package lists were updated, and there were no pending upgrades (0 upgraded, 0 newly installed, 0 to remove, and 0 not upgraded).

Evidence:

```
castor909@mtu:~$ sudo apt update && sudo apt upgrade
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://es.archive.ubuntu.com/ubuntu noble InRelease
Hit:3 https://deb.nodesource.com/node_24.x nodistro InRelease
Hit:4 http://es.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:5 http://es.archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
  linux-headers-6.8.0-90 linux-headers-6.8.0-90-generic linux-image-6.8.0-90-generic linux-modules-6.8.0-90-generic linux-modules-extra-6.8.0-90-generic
  linux-tools-6.8.0-90 linux-tools-6.8.0-90-generic
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove, 0 not upgraded.
```

1.2 LDAP packages installation

What I did: Installed OpenLDAP server and LDAP client utilities, then verified both packages are present in the system.

Commands used: `sudo apt install -y slapd ldap-utils sudo dpkg -l | grep -E "slapd|ldap-utils"`

Result: Both required packages are installed (`ldap-utils` and `slapd` , version `2.6.10+dfsg-0ubuntu0.24.04.1` , amd64).

Evidence:

```
castor909@mus:~$ sudo dpkg -l | grep -E "slapd|ldap-utils"
ii  ldap-utils      2.6.10+dfsg-0ubuntu0.24.04.1      amd64      OpenLDAP utilities
ii  slapd           2.6.10+dfsg-0ubuntu0.24.04.1      amd64      OpenLDAP server
```

1.3 Service activation and verification

What I did: Enabled `slapd` for automatic startup and verified runtime status with `systemd`.

Commands used: `sudo systemctl enable slapd sudo systemctl status slapd --no-pager sudo systemctl is-enabled slapd`

Result: `slapd.service` is active and running. Ubuntu reports a SysV compatibility redirection for enablement (`systemd-sysv-install`) because `slapd` is managed through generated service wrappers; this behavior is expected.

Evidence:

```
castor909@mus:~$ sudo systemctl status slapd --no-pager
■ slapd.service - LSB: OpenLDAP standalone server (Lightweight Directory Access Protocol)
   Loaded: loaded (/etc/init.d/slapd; generated)
   Drop-In: /usr/lib/systemd/system/slapd.service.d
            └─slapd-remain-after-exit.conf
   Active: active (running) since Sun 2026-04-19 21:14:17 UTC; 21min ago
     Docs: man:systemd-sysv-generator(8)
    Tasks: 3 (limit: 2263)
  Memory: 3.4M (peak: 4.3M)
     CPU: 14ms
   CGroup: /system.slice/slapd.service
            └─33930 /usr/sbin/slapd -h "ldap:/// ldapi:///" -g openldap -u openldap -F /etc/ldap/slapd.d

Apr 19 21:14:17 mus systemd[1]: Starting slapd.service - LSB: OpenLDAP standalone server (Lightweight Directory Access Protocol).
Apr 19 21:14:17 mus slapd[33923]: * Starting OpenLDAP slapd
Apr 19 21:14:17 mus slapd[33929]: o(=) $OpenLDAP: slapd 2.6.10+dfsg-0ubuntu0.24.04.1 (Sep 23 2025 16:26:39) $
                        Ubuntu Developers <ubuntu-devel-discuss@lists.ubuntu.com>
Apr 19 21:14:17 mus slapd[33930]: slapd starting
Apr 19 21:14:17 mus slapd[33923]: ...done.
Apr 19 21:14:17 mus systemd[1]: Started slapd.service - LSB: OpenLDAP standalone server (Lightweight Directory Access Protocol).
castor909@mus:~$ sudo systemctl is-enabled slapd
slapd.service is not a native service, redirecting to systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install is-enabled slapd
enabled
```

1.4 Firewall rules

What I did: Verified UFW status and confirmed inbound rules for SSH and LDAP.

Commands used: `sudo ufw status verbose`

Result: UFW is active with default deny for incoming traffic, and explicit allow rules are present for `22/tcp` (SSH) and `389/tcp` (LDAP), including IPv6 entries.

Evidence:

```
castor909@mus: $ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip
```

To	Action	From
22/tcp	ALLOW IN	Anywhere
389/tcp	ALLOW IN	Anywhere
22/tcp (v6)	ALLOW IN	Anywhere (v6)
389/tcp (v6)	ALLOW IN	Anywhere (v6)

2. LDAP base configuration and initial directory tree

2.0 Base DN plan

Chosen domain: dawserver.com

Base DN: dc=dawserver,dc=com

Organization name: dawserver

2.1 slapd reconfiguration

What I did: Reconfigured `slapd` with `dpkg-reconfigure slapd` and defined the LDAP naming context for the practice.

Base DN: dc=dawserver,dc=com

Result: Reconfiguration completed successfully using MDB backend and a new admin password; LDAP service remained available after configuration.

Evidence:

```
Backing up /etc/ldap/slapd.d in /var/backups/slapd-2.6.10+dfsg-0ubuntu0.24.04.1... done.
Moving old database directory to /var/backups:
- directory unknown... done.
Creating initial configuration... done.
Creating LDAP directory... done.
```

2.2 Base LDIF

LDIF content:

```
dn: dc=dawserver,dc=com
objectClass: top
objectClass: dcObject
objectClass: organization
o: dawserver
dc: dawserver

dn: ou=usuarios,dc=dawserver,dc=com
objectClass: organizationalUnit
ou: usuarios

dn: ou=grupos,dc=dawserver,dc=com
objectClass: organizationalUnit
ou: grupos
```

Import command: `ldapadd -x -D "cn=admin,dc=dawserver,dc=com" -W -f base.ldif`

After the first attempt returned `Already exists (68)` for the base DN, the OU entries were imported separately:

`ldapadd -x -D "cn=admin,dc=dawserver,dc=com" -W -f ous.ldif`

Result: The base domain entry already existed after reconfiguration, and both organizational units were created successfully (`ou=usuarios` and `ou=grupos`).

Evidence:

```

castor909@mus:~$ cat > base.ldif <<'EOF'
> dn: dc=dawserver,dc=com
> objectClass: top
> objectClass: dcObject
> objectClass: organization
> o: dawserver
> dc: dawserver
>
> dn: ou=usuarios,dc=dawserver,dc=com
> objectClass: organizationalUnit
> ou:usuarios
>
> dn: ou=grupos,dc=dawserver,dc=com
> objectClass: organizationalUnit
> ou:grupos
> EOF
castor909@mus:~$ ldapadd -x -D "cn=admin,dc=dawserver,dc=com" -W -f base.ldif
Enter LDAP Password:
adding new entry "dc=dawserver,dc=com"
ldap_add: Already exists (68)

castor909@mus:~$ ldapsearch -x LLL -b "dc=dawserver,dc=com"
# extended LDIF
#
# LDAPv3
# base <dc=dawserver,dc=com> with scope subtree
# filter: (objectclass=*)
# requesting: LLL
#
# dawserver.com
dn: dc=dawserver,dc=com

# search result
search: 2
result: 0 Success

# numResponses: 2
# numEntries: 1

```

2.3 Directory verification

Search command: `ldapsearch -x -LLL -b "dc=dawserver,dc=com" ldapsearch -x -LLL -b "dc=dawserver,dc=com" "(ou=usuarios)" ldapsearch -x -LLL -b "dc=dawserver,dc=com" "(ou=grupos)"`

Result: LDAP search confirms the directory tree contains the base DN plus both OUs. Filtered searches by `ou` also return the expected entries.

Evidence:

```

castor909@mus:~$ ldapadd -x -D "cn=admin,dc=dawserver,dc=com" -W -f ous.ldif
Enter LDAP Password:
adding new entry "ou=usuarios,dc=dawserver,dc=com"

adding new entry "ou=grupos,dc=dawserver,dc=com"

castor909@mus:~$ ldapsearch -x -LLL -b "dc=dawserver,dc=com"
dn: dc=dawserver,dc=com
objectClass: top
objectClass: dcObject
objectClass: organization
o: dawserver
dc: dawserver

dn: ou=usuarios,dc=dawserver,dc=com
objectClass: organizationalUnit
ou: usuarios

dn: ou=grupos,dc=dawserver,dc=com
objectClass: organizationalUnit
ou: grupos

castor909@mus:~$ ldapsearch -x -LLL -b "dc=dawserver,dc=com" "(ou=usuarios)"
dn: ou=usuarios,dc=dawserver,dc=com
objectClass: organizationalUnit
ou: usuarios

castor909@mus:~$ ldapsearch -x -LLL -b "dc=dawserver,dc=com" "(ou=grupos)"
dn: ou=grupos,dc=dawserver,dc=com
objectClass: organizationalUnit
ou: grupos

```

3. Users and groups

3.0 User and group plan

User 1: usuario1

User 2: usuario2

User 3: usuario3

Group: webusers

3.1 User entries

What I did: Imported LDAP user entries for three practice accounts and verified that the entries exist in the directory.

LDIF content:

```
dn: uid=usuario1,ou=usuarios,dc=dawserver,dc=com
objectClass: inetOrgPerson
objectClass: posixAccount
objectClass: shadowAccount
cn: Usuario Uno
sn: Uno
uid: usuario1
uidNumber: 10001
gidNumber: 10000
homeDirectory: /home/usuario1
loginShell: /bin/bash
mail: usuario1@dawserver.com
userPassword: usuario1pass

dn: uid=usuario2,ou=usuarios,dc=dawserver,dc=com
objectClass: inetOrgPerson
objectClass: posixAccount
objectClass: shadowAccount
cn: Usuario Dos
sn: Dos
uid: usuario2
uidNumber: 10002
gidNumber: 10000
homeDirectory: /home/usuario2
loginShell: /bin/bash
mail: usuario2@dawserver.com
userPassword: usuario2pass

dn: uid=usuario3,ou=usuarios,dc=dawserver,dc=com
objectClass: inetOrgPerson
objectClass: posixAccount
objectClass: shadowAccount
cn: Usuario Tres
sn: Tres
uid: usuario3
uidNumber: 10003
gidNumber: 10000
homeDirectory: /home/usuario3
loginShell: /bin/bash
mail: usuario3@dawserver.com
userPassword: usuario3pass
```

Result: ldapadd reported `Already exists (68)` on re-import, which confirms the

`usuario2` correctly stored with all required attributes.

Evidence:

```
castor909@mus:~$ ldapadd -x -D "cn=admin,dc=dawserver,dc=com" -W -f users.ldif
Enter LDAP Password:
adding new entry "uid=usuario1,ou=usuarios,dc=dawserver,dc=com"
ldap_add: Already exists (68)
```

3.2 Group entry

What I did: Imported the practice group and verified that it contains the expected user memberships.

LDIF content:

```
dn: cn=webusers,ou=grupos,dc=dawserver,dc=com
objectClass: posixGroup
cn: webusers
gidNumber: 10000
memberUid: usuario1
memberUid: usuario2
memberUid: usuario3
```

Result: `ldapadd` reported `Already exists (68)` on re-import, which confirms the group already exists. The search output shows the `webusers` group with `gidNumber` and all three `memberUid` values.

Evidence:

```
castor909@mus:~$ ldapadd -x -D "cn=admin,dc=dawserver,dc=com" -W -f group.ldif
Enter LDAP Password:
adding new entry "cn=webusers,ou=grupos,dc=dawserver,dc=com"
ldap_add: Already exists (68)
```

3.3 LDAP verification queries

Search command: `ldapsrch -x -LLL -b "ou=usuarios,dc=dawserver,dc=com" "(uid=usuario1)"`
`ldapsrch -x -LLL -b "ou=usuarios,dc=dawserver,dc=com" "(cn=Usuario Dos)"`
`ldapsrch -x -LLL -b "ou=grupos,dc=dawserver,dc=com" "(cn=webusers)"`

Result: The directory returns the expected user and group entries. `usuario1` and `usuario2` are visible with their attributes, and `webusers` contains `usuario1`, `usuario2`, and `usuario3` as members.

Evidence:


```

castor909@mus:~$ ldapsearch -x -LLL -b "ou=usuarios,dc=dawserver,dc=com" "(uid=usuario1)"
dn: uid=usuario1,ou=usuarios,dc=dawserver,dc=com
objectClass: inetOrgPerson
objectClass: posixAccount
objectClass: shadowAccount
cn: Usuario Uno
sn: Uno
uid: usuario1
uidNumber: 10001
gidNumber: 10000
homeDirectory: /home/usuario1
loginShell: /bin/bash
mail: usuario1@dawserver.com

castor909@mus:~$ ldapsearch -x -LLL -b "ou=usuarios,dc=dawserver,dc=com" "(cn=Usuario Dos)"
dn: uid=usuario2,ou=usuarios,dc=dawserver,dc=com
objectClass: inetOrgPerson
objectClass: posixAccount
objectClass: shadowAccount
cn: Usuario Dos
sn: Dos
uid: usuario2
uidNumber: 10002
gidNumber: 10000
homeDirectory: /home/usuario2
loginShell: /bin/bash
mail: usuario2@dawserver.com

castor909@mus:~$ ldapsearch -x -LLL -b "ou=grupos,dc=dawserver,dc=com" "(cn=webusers)"
dn: cn=webusers,ou=grupos,dc=dawserver,dc=com
objectClass: posixGroup
cn: webusers
gidNumber: 10000
memberUid: usuario1
memberUid: usuario2
memberUid: usuario3

```

4. Apache integration with LDAP authentication

4.0 Apache plan

Protected path: /var/www/html/privado

LDAP URL: ldap://127.0.0.1:389/ou=usuarios,dc=dawserver,dc=com?uid?sub?(objectClass=inetOrgPerson)

Apache modules enabled: ldap, authnz_ldap

4.1 Protected directory

What I did: Created and protected the /var/www/html/privado directory to require LDAP authentication.

Result: Apache accepted the LDAP auth configuration. `apache2ctl configtest` returns only an FQDN warning (non-blocking).

Evidence:

```

castor909@mus:~$ sudo apache2ctl configtest
AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 2a0c:5a86:d18a:f200:a00:27ff:fed0:7e81. Set the 'ServerName' directive globally to suppress this message
Syntax OK

```

4.2 Apache LDAP configuration

Config snippet:

```
<Directory /var/www/html/privado>
    AuthType Basic
    AuthName "Acceso privado LDAP"
    AuthBasicProvider ldap
    AuthLDAPURL "ldap://127.0.0.1:389/ou=usuarios,dc=dawserver,dc=com?uid?sub?(objectClass=inetOrgPerson)"
    AuthLDAPBindDN "cn=admin,dc=dawserver,dc=com"
    AuthLDAPBindPassword "[REDACTED]"
    Require valid-user
</Directory>
```

Result: Apache LDAP directives were applied correctly for the protected directory.

Evidence:

```
castor909@mus:~$ cat <<'EOF' | sudo tee /etc/apache2/conf-available/ldap-privado.conf
<Directory /var/www/html/privado>
    AuthType Basic
    AuthName "Acceso privado LDAP"
    AuthBasicProvider ldap
    AuthLDAPURL "ldap://127.0.0.1:389/ou=usuarios,dc=dawserver,dc=com?uid?sub?(objectClass=inetOrgPerson)"
    AuthLDAPBindDN "cn=admin,dc=dawserver,dc=com"
    AuthLDAPBindPassword "arinsea"
    Require valid-user
</Directory>
EOF
```

```
castor909@mus:~$ curl -i http://localhost/privado/
HTTP/1.1 401 Unauthorized
Date: Mon, 20 Apr 2026 00:29:31 GMT
Server: Apache/2.4.58 (Ubuntu)
WWW-Authenticate: Basic realm="Acceso privado LDAP"
Content-Length: 456
Content-Type: text/html; charset=iso-8859-1

<!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2.0//EN">
<html><head>
<title>401 Unauthorized</title>
</head><body>
<h1>Unauthorized</h1>
<p>This server could not verify that you
are authorized to access the document
requested. Either you supplied the wrong
credentials (e.g., bad password), or your
browser doesn't understand how to supply
the credentials required.</p>
<hr>
<address>Apache/2.4.58 (Ubuntu) Server at localhost Port 80</address>
</body></html>
```

4.3 Access tests

Denied without credentials: `curl -i http://localhost/privado/` returns `HTTP/1.1 401 Unauthorized`.

Allowed with LDAP user: `curl -i -u usuario1:usuario1pass http://localhost/privado/` returns `HTTP/1.1 200 OK` and serves the private page content.

Evidence:

```
castor909@mus:~$ curl -i -u usuario1:usuario1pass http://localhost/privado/
HTTP/1.1 200 OK
Date: Mon, 20 Apr 2026 00:30:33 GMT
Server: Apache/2.4.58 (Ubuntu)
Last-Modified: Mon, 20 Apr 2026 00:21:46 GMT
ETag: "4f-64fd948ac8884"
Accept-Ranges: bytes
Content-Length: 79
Vary: Accept-Encoding
Content-Type: text/html

<h1>Zona privada LDAP</h1>
<p>Si ves esto, la autentificacion LDAP funciona.</p>
```

5. Client-side verification and traffic capture

5.0 Client verification plan

LDAP client command: `ldapwhoami -x -D`

`"uid=usuario1,ou=usuarios,dc=dawserver,dc=com" -W`

Browser test location: Arch Linux host (outside the VM) using curl against `http://192.168.1.140/privado/`

Capture tool: tcpdump or Wireshark on the Ubuntu Server VM (interface connected to 192.168.1.140)

5.1 LDAP authentication test

Command: `ldapwhoami -x -D`

`"uid=usuario1,ou=usuarios,dc=dawserver,dc=com" -W`

Result: Successful LDAP bind. The server returned the authenticated identity:
`dn:uid=usuario1,ou=usuarios,dc=dawserver,dc=com`

Evidence:

```
castor909@mus:~$ ldapwhoami -x -D "uid=usuario1,ou=usuarios,dc=dawserver,dc=com" -W
Enter LDAP Password:
dn:uid=usuario1,ou=usuarios,dc=dawserver,dc=com
```

5.2 Browser access from client

Result: From the Arch Linux host, client-to-server access behaves as expected.

- Without credentials: `HTTP/1.1 401 Unauthorized` for `GET /privado/`.
- With LDAP credentials (`usuario1:usuario1pass`): `HTTP/1.1 200 OK` and private page content returned.

This confirms remote Apache LDAP authentication works from another machine, not only locally on the VM.

Evidence:

```
~/repos/desplegament-d-aplicacions-web/practica-12-ldap main*
> curl -4 -v --connect-timeout 5 --max-time 10 http://192.168.1.140/privado/
>
* Request completely sent off
< HTTP/1.1 401 Unauthorized
< Date: Mon, 20 Apr 2026 01:05:34 GMT
< Server: Apache/2.4.58 (Ubuntu)
< WWW-Authenticate: Basic realm="Acceso privado LDAP"
< Content-Length: 460
< Content-Type: text/html; charset=iso-8859-1
<
<!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2.0//EN">
<html><head>
<title>401 Unauthorized</title>
</head><body>
<h1>Unauthorized</h1>
<p>This server could not verify that you
are authorized to access the document
requested. Either you supplied the wrong
credentials (e.g., bad password), or your
browser doesn't understand how to supply
the credentials required.</p>
<hr>
<address>Apache/2.4.58 (Ubuntu) Server at 192.168.1.140 Port 80</address>
</body></html>
* Connection #0 to host 192.168.1.140:80 left intact

~/repos/desplegament-d-aplicacions-web/practica-12-ldap main*
● > curl -4 -v --connect-timeout 5 --max-time 10 -u usuario1:usuario1pass http://192.168.1.140/privado/
* Trying 192.168.1.140:80...
* Established connection to 192.168.1.140 (192.168.1.140 port 80) from 192.168.1.133 port 38828
* using HTTP/1.x
* Server auth using Basic with user 'usuario1'
> GET /privado/ HTTP/1.1
> Host: 192.168.1.140
> Authorization: Basic dXN1YXJpbzE6dXN1YXJpbzFwYXNz
> User-Agent: curl/8.19.0
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< Date: Mon, 20 Apr 2026 01:05:53 GMT
< Server: Apache/2.4.58 (Ubuntu)
< Last-Modified: Mon, 20 Apr 2026 00:21:46 GMT
< ETag: "4f-64fd948ac8884"
< Accept-Ranges: bytes
< Content-Length: 79
< Vary: Accept-Encoding
< Content-Type: text/html
<
<h1>Zona privada LDAP</h1>
<p>Si ves esto, la autenticacion LDAP funciona.</p>
* Connection #0 to host 192.168.1.140:80 left intact

~/repos/desplegament-d-aplicacions-web/practica-12-ldap main*
```

5.3 Traffic capture

Tool used: `tcpdump` on the Ubuntu Server VM (`sudo tcpdump -i any -n -vv "tcp port 80 or tcp port 389"`).

What the capture proves: The capture shows real client-to-server HTTP traffic from `192.168.1.133` to `192.168.1.140:80`, including `GET /privado/` and `HTTP/1.1 200 OK`. This confirms the authenticated request path from external client to Apache. In this test run, the screenshot clearly documents web authentication flow over HTTP; LDAP bind verification is already proven in section 5.1 (`ldapwhoami`).

Evidence:

```

192.168.1.133.55558 > 192.168.1.140.80: Flags [I.], cksum 0xf46c (correct), seq 87, ack 679, win 63, options [nop,nop,TS val 301841070 ecr 874693052], length 0
01:10:56.496463 enp0s3 In IP (tos 0x0, ttl 64, id 64813, offset 0, flags [DF], proto TCP (6), length 60)
192.168.1.133.55562 > 192.168.1.140.80: Flags [S], cksum 0xaf08 (correct), seq 1296411513, win 64240, options [mss 1460,sackOK,TS val 2507846157 ecr 0,nop,wscale 10], length 0
01:10:56.496522 enp0s3 Out IP (tos 0x0, ttl 64, id 0, offset 0, flags [DF], proto TCP (6), length 60)
192.168.1.140.80 > 192.168.1.133.55562: Flags [S.], cksum 0xb490 (incorrect -> 0x9e2c), seq 1399636965, ack 1296411514, win 65160, options [mss 1460,sackOK,TS val 874693057 ecr 2507846157,nop,wscale 7], length 0
01:10:56.496841 enp0s3 In IP (tos 0x0, ttl 64, id 64814, offset 0, flags [DF], proto TCP (6), length 52)
192.168.1.133.55562 > 192.168.1.140.80: Flags [I.], cksum 0xcb42 (correct), seq 1, ack 1, win 63, options [nop,nop,TS val 2507846157 ecr 874693057], length 0
01:10:56.496841 enp0s3 In IP (tos 0x0, ttl 64, id 64815, offset 0, flags [DF], proto TCP (6), length 188)
192.168.1.133.55562 > 192.168.1.140.80: Flags [P.], cksum 0x23dc (correct), seq 1:137, ack 1, win 63, options [nop,nop,TS val 2507846157 ecr 874693057], length 136: HTTP, length: 136
GET /privado/ HTTP/1.1
Host: 192.168.1.140
Authorization: Basic dXN1YXJpbzE6dXN1YXJpbzFuYXNz
User-Agent: curl/8.19.0
Accept: */*

01:10:56.496885 enp0s3 Out IP (tos 0x0, ttl 64, id 18390, offset 0, flags [DF], proto TCP (6), length 52)
192.168.1.140.80 > 192.168.1.133.55562: Flags [I.], cksum 0x8488 (incorrect -> 0xc8fc), seq 1, ack 137, win 508, options [nop,nop,TS val 874693058 ecr 2507846157], length 0
01:10:56.497263 enp0s3 Out IP (tos 0x0, ttl 64, id 18391, offset 0, flags [DF], proto TCP (6), length 381)
192.168.1.140.80 > 192.168.1.133.55562: Flags [P.], cksum 0x85d1 (incorrect -> 0x6b42), seq 1:330, ack 137, win 508, options [nop,nop,TS val 874693058 ecr 2507846157], length 329: HTTP, length: 329
HTTP/1.1 200 OK
Date: Mon, 20 Apr 2026 01:10:56 GMT
Server: Apache/2.4.58 (Ubuntu)
Last-Modified: Mon, 20 Apr 2026 00:21:46 GMT
ETag: "4f-64f40940ac8804"
Accept-Ranges: bytes
Content-Length: 79
Vary: Accept-Encoding
Content-Type: text/html

<h1>Zona privada LDAP</h1>
<p>Si ves esto, la autenticacion LDAP funciona.</p>
01:10:56.497345 enp0s3 In IP (tos 0x0, ttl 64, id 64816, offset 0, flags [DF], proto TCP (6), length 52)
192.168.1.133.55562 > 192.168.1.140.80: Flags [I.], cksum 0xc96f (correct), seq 137, ack 330, win 63, options [nop,nop,TS val 2507846158 ecr 874693058], length 0
01:10:56.497443 enp0s3 In IP (tos 0x0, ttl 64, id 64817, offset 0, flags [DF], proto TCP (6), length 52)
192.168.1.133.55562 > 192.168.1.140.80: Flags [F.], cksum 0xc96e (correct), seq 137, ack 330, win 63, options [nop,nop,TS val 2507846158 ecr 874693058], length 0
01:10:56.497522 enp0s3 Out IP (tos 0x0, ttl 64, id 18392, offset 0, flags [DF], proto TCP (6), length 52)
192.168.1.140.80 > 192.168.1.133.55562: Flags [F.], cksum 0x8488 (incorrect -> 0xc7b0), seq 330, ack 138, win 508, options [nop,nop,TS val 874693058 ecr 2507846158], length 0
01:10:56.497574 enp0s3 In IP (tos 0x0, ttl 64, id 64818, offset 0, flags [DF], proto TCP (6), length 52)
192.168.1.133.55562 > 192.168.1.140.80: Flags [I.], cksum 0xc96d (correct), seq 138, ack 331, win 63, options [nop,nop,TS val 2507846158 ecr 874693058], length 0

```

Bonus (optional)

Secure LDAP or phpLDAPadmin

What I did: No bonus feature was added in this submission.

Result: The mandatory tasks were prioritized and completed in a stable state (LDAP directory, Apache integration, client verification, and traffic capture).

Evidence: Not applicable.

Conclusion

This practice deployed a complete LDAP-based authentication environment on Ubuntu Server and validated it end-to-end from an external client machine.

The OpenLDAP service was installed and configured with base DN

`dc=dawserver,dc=com`, organizational units (`ou=usuarios`, `ou=grupos`), three users, and one group (`webusers`). Directory consistency was verified with multiple `ldapsearch` queries.

Apache was integrated with LDAP authentication for a protected directory (`/var/www/html/privado`) using `mod_ldap` and `mod_authnz_ldap`. Access behavior was validated both locally and remotely: unauthenticated requests returned `401 Unauthorized`, while valid LDAP credentials returned `200 OK`.

Final client-side validation included `ldapwhoami` successful bind and packet capture evidence with `tcpdump`, confirming the HTTP authentication flow between host and VM. The resulting setup demonstrates centralized credential

Appendix

Useful commands

- `apt update && apt upgrade`
- `apt install slapd ldap-utils`
- `ldapsearch`
- `ldapadd`
- `ldapwhoami`
- Apache restart and status commands

Notes

- Report title and student full name are already filled for this submission.
- Keep the report in the same order as the assignment.
- Add concise explanations below each screenshot.

Screenshot inventory

Save the screenshots inside the `captures` folder using these filenames so they can be embedded into the markdown report later.

- `1.1-update-upgrade.png` - apt update and apt upgrade output.
- `1.2-install-packages.png` - slapd and ldap-utils installation plus package verification.
- `1.3-slapd-status.png` - systemctl enable/status/is-enabled output for slapd.
- `1.4-ufw-status.png` - UFW verbose status showing SSH and LDAP rules.
- `2.1-slapd-reconfigure.png` - slapd reconfiguration wizard completed.
- `2.2-base-ldif-import.png` - base LDIF and OU import output.
- `2.3-base-search.png` - ldapsearch showing the base tree and OU filters.
- `3.1-users-import.png` - user LDIF import output.
- `3.2-group-import.png` - group LDIF import output.
- `3.3-ldapsearch-users-group.png` - filtered ldapsearch results for users and group.
- `4.1-apache-config.png` - Apache configtest output.
- `4.2-apache-ldap-config.png` - Apache LDAP auth configuration file content.
- `4.2-apache-denied.png` - browser access denied without credentials.
- `4.3-apache-allowed.png` - browser access allowed with LDAP credentials.
- `5.1-ldapwhoami.png` - client authentication test.
- `5.2-browser-client.png` - browser access from client machine.
- `5.3-capture.png` - Wireshark or tcpdump capture.